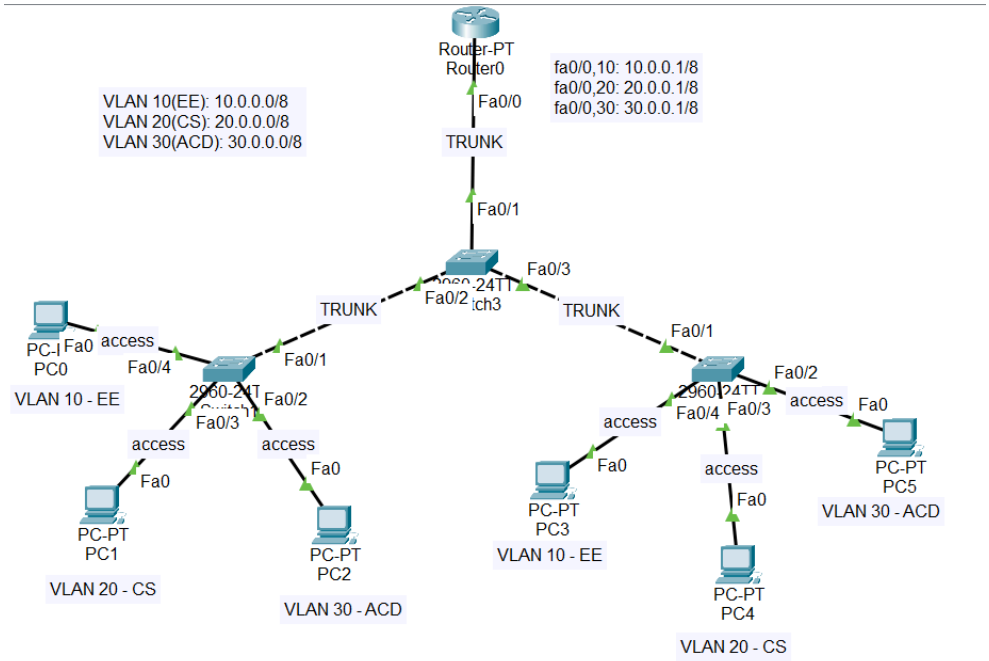
LAB 12

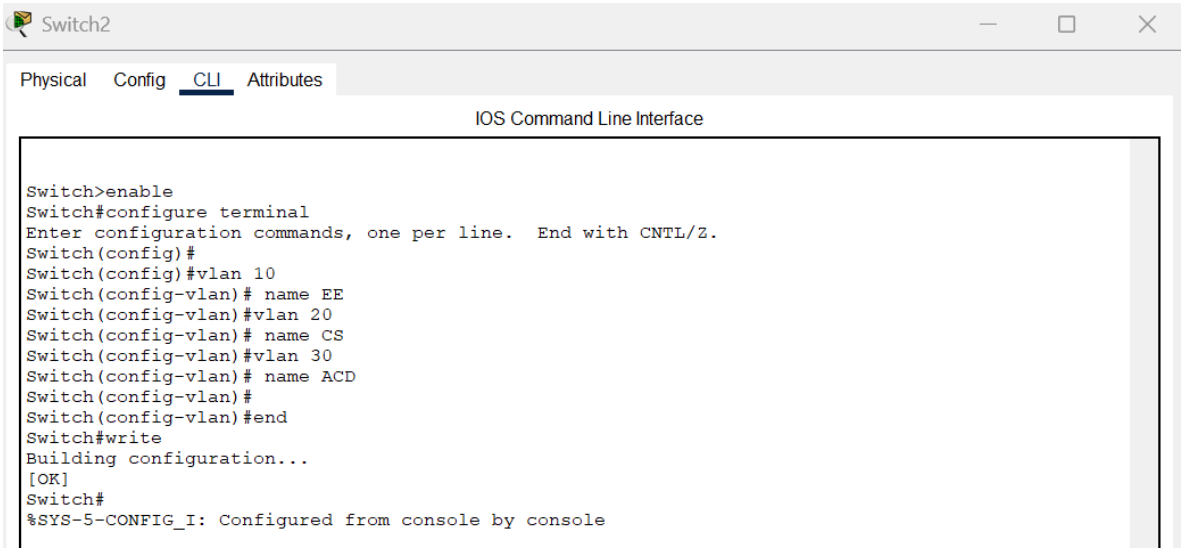
23K-0057

Task 1:

Topology:



Creating VLANS on Switches:




```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name EE
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name CS
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name ACD
Switch(config-vlan)#
Switch(config-vlan)#end
Switch#write
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name EE
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name CS
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name ACD
Switch(config-vlan)#
Switch(config-vlan)#end
Switch#write
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name EE
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name CS
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name ACD
Switch(config-vlan)#
Switch(config-vlan)#end
Switch#write
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuring access ports for PCs:

Left Switch:

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#en
Switch#config t
Enter configuration commands, one per line.  End with
CNTL/Z.
Switch(config)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#int fa 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#
```

Right Switch:

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#! PC5 - VLAN 30 on Fa0/2
Switch(config)#interface fastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#! PC4 - VLAN 20 on Fa0/3
Switch(config)#interface fastEthernet0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#! PC3 - VLAN 10 on Fa0/4
Switch(config)#interface fastEthernet0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#write
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuring trunk links between switches and router:

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
```

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch con0 is now available



Press RETURN to get started.



Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#fa0/1
               ^
% Invalid input detected at '^' marker.

Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
```

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch con0 is now available



Press RETURN to get started.



Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#witchport mode trunk
                 ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchportmode trun
                            ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode trunk
```

Router Configuration:

Router0

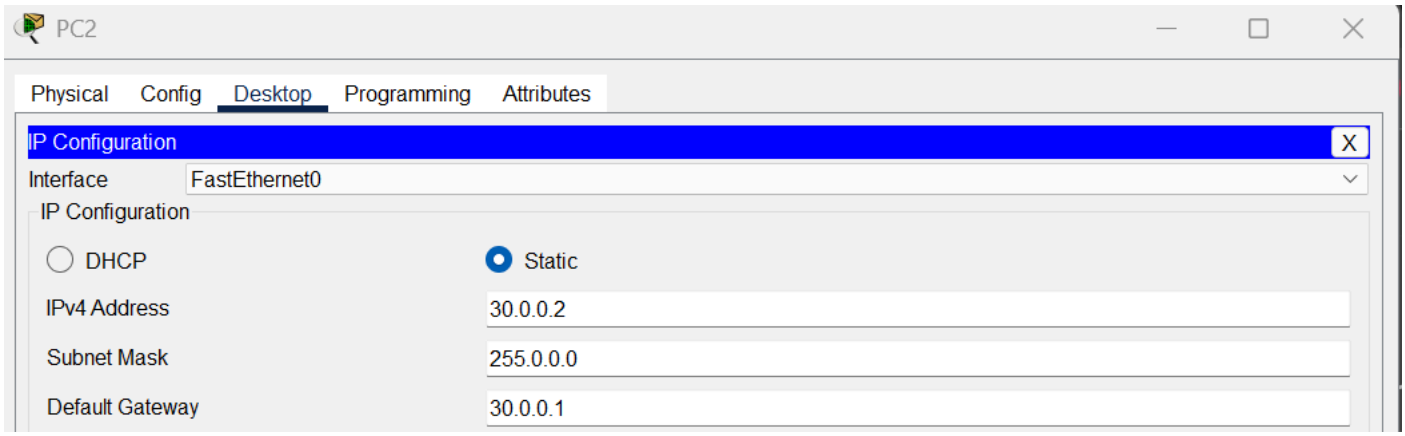
Physical Config CLI Attributes

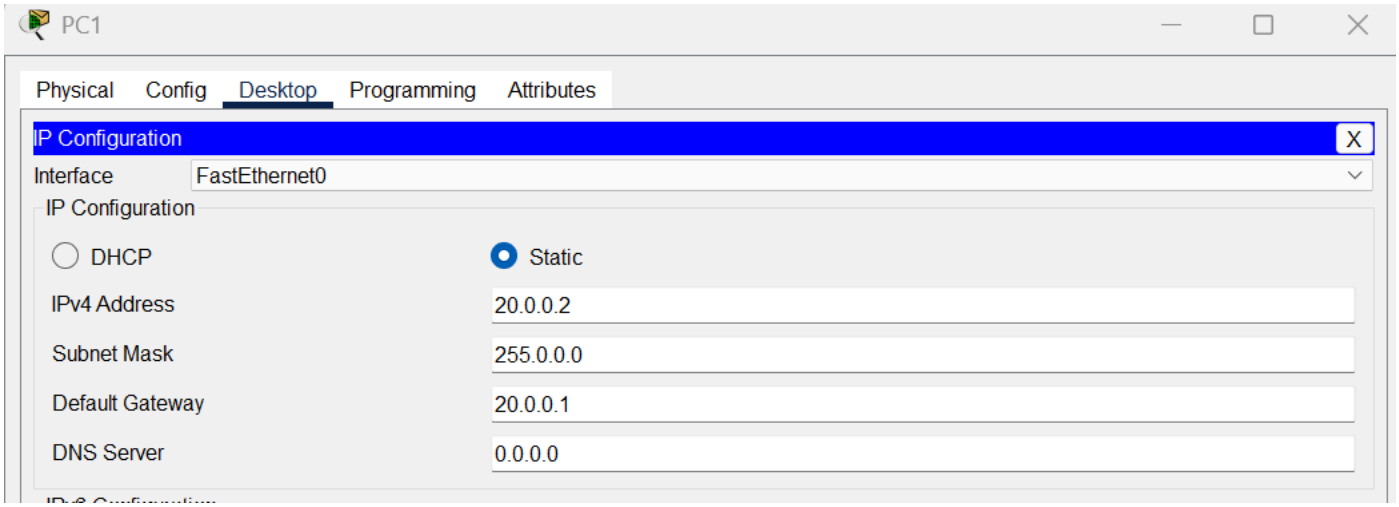
IOS Command Line Interface

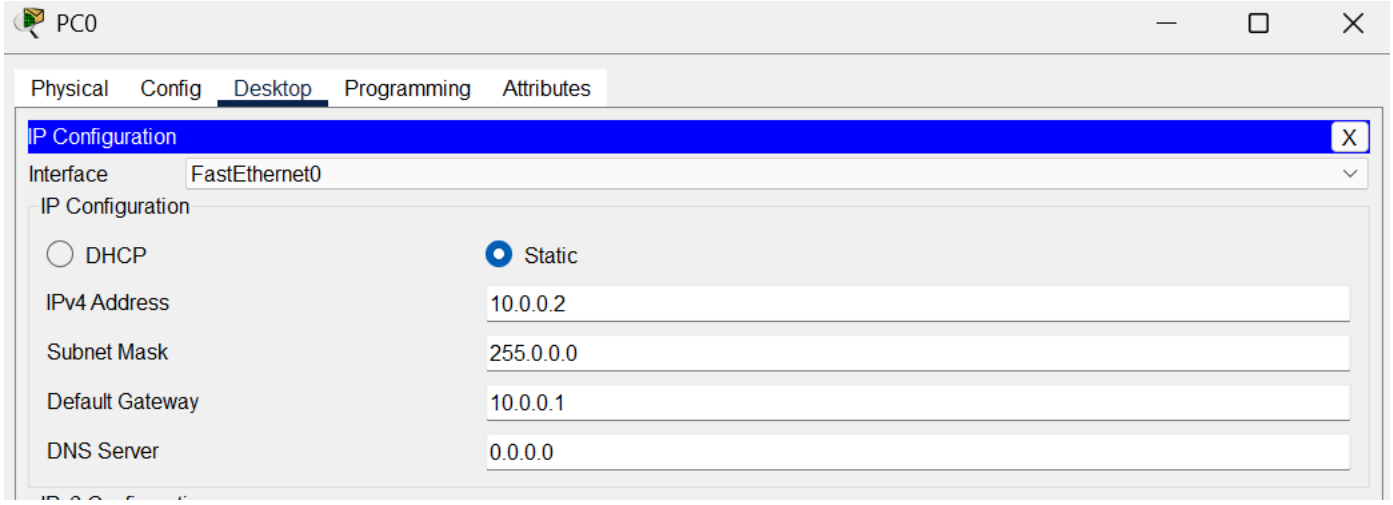
```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#! Subinterface for VLAN 10 (EE)
Router(config)#interface fastEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 10.0.0.1 255.0.0.0
Router(config-subif)#exit
Router(config)#
Router(config)#! Subinterface for VLAN 20 (CS)
Router(config)#interface fastEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 20.0.0.1 255.0.0.0
Router(config-subif)#exit
Router(config)#
Router(config)#! Subinterface for VLAN 30 (ACD)
Router(config)#interface fastEthernet0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 30.0.0.1 255.0.0.0
Router(config-subif)#exit
Router(config)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up
```

Copy Paste

PCs Configurations:







PC5

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 30.0.0.3

Subnet Mask: 255.0.0.0

Default Gateway: 30.0.0.1

DNS Server: 0.0.0.0

IPv6 Configuration

PC3

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 10.0.0.3

Subnet Mask: 255.0.0.0

Default Gateway: 10.0.0.1

DNS Server: 0.0.0.0

PC4

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 20.0.0.3

Subnet Mask: 255.0.0.0

Default Gateway: 20.0.0.1

DNS Server: 0.0.0.0

Verification:

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6,
Fa0/7
                                               Fa0/8, Fa0/9, Fa0/10,
Fa0/11
                                               Fa0/12, Fa0/13, Fa0/14,
Fa0/15
                                               Fa0/16, Fa0/17, Fa0/18,
Fa0/19
                                               Fa0/20, Fa0/21, Fa0/22,
Fa0/23
                                               Fa0/24, Gig0/1, Gig0/2
10   EE                                active
20   CS                                active
30   ACD                               active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       auto         n-802.1q       trunking      1
Fa0/3       auto         n-802.1q       trunking      1
```

Copy Paste

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

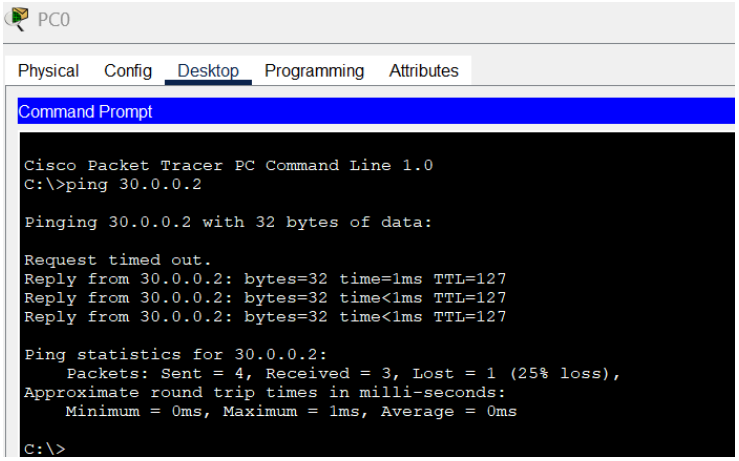
```
Switch con0 is now available



Press RETURN to get started.



Switch>en
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                               Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                               Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                               Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                               Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                               Gig0/1, Gig0/2
10   EE                                active    Fa0/4
20   CS                                active    Fa0/3
30   ACD                               active    Fa0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
```

Similarly checked at switch 2

```
Router#en
Router#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  up                    up
FastEthernet0/0.10     10.0.0.1        YES manual up                    up
FastEthernet0/0.20     20.0.0.1        YES manual up                    up
FastEthernet0/0.30     30.0.0.1        YES manual up                    up
FastEthernet1/0        unassigned      YES unset  administratively down down
Serial2/0              unassigned      YES unset  administratively down down
Serial3/0              unassigned      YES unset  administratively down down
FastEthernet4/0        unassigned      YES unset  administratively down down
FastEthernet5/0        unassigned      YES unset  administratively down down
Router#
```

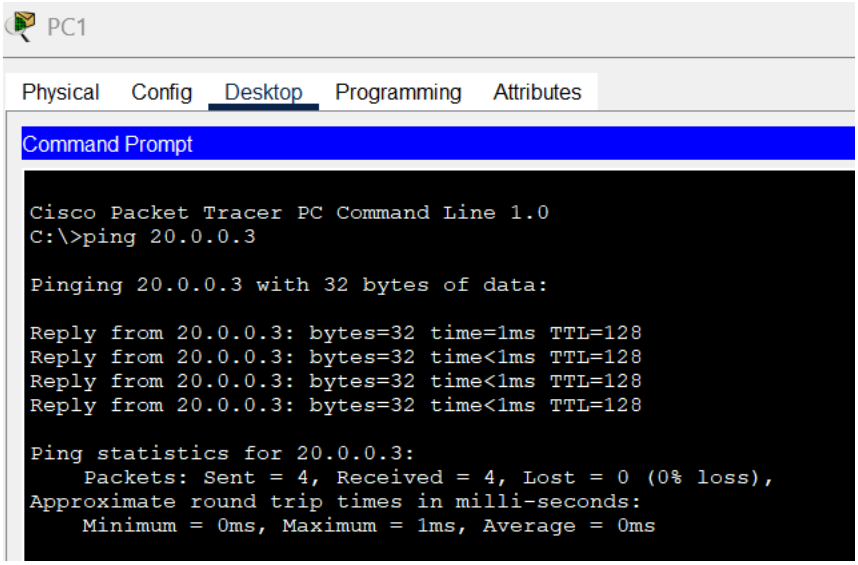
Same VLAN communication:

PC0 TO PC2:



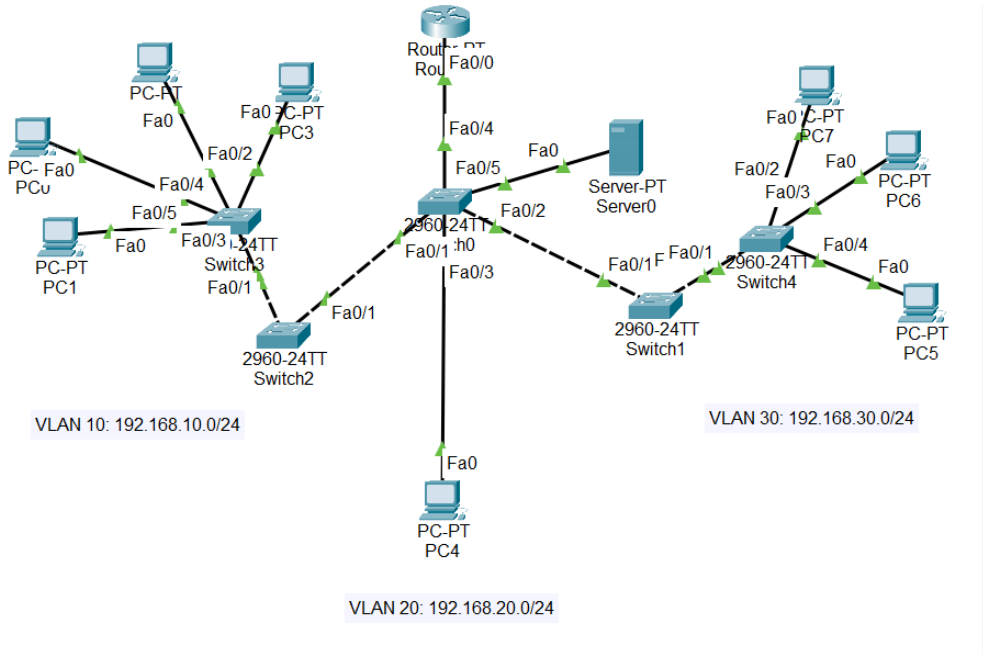
Inter VLAN communication:

PC1 TO PC4:



Task 2:

Topology:



Configurations:

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Cisco Internetwork Operating System Software
IOS (tm) PT1000 Software (PT1000-I-M), Version 12.2(28), RELEASE SOFTWARE (fc5)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 27-Apr-04 19:01 by miwang

PT 1001 (PTSC2005) processor (revision 0x200) with 60416K/5120K bytes of memory
.
Processor board ID PT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#config
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no shut
```

Copy Paste

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface fa0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface fa0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface fa0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.30, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Copy Paste

VLAN Creation on switches:

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up


Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)# name LEFT_VLAN
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name MID_VLAN
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name RIGHT_VLAN
Switch(config-vlan)#exit
Switch(config)#end
Switch#wr
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Hardware Board Revision Number  : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up


Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)# name LEFT_VLAN
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name MID_VLAN
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name RIGHT_VLAN
Switch(config-vlan)#exit
Switch(config)#end
Switch#wr
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Top

Similarly done for all other switches.

Switch Access Ports Configurations:

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started.



Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
```

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
```

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   LEFT_VLAN                        active
20   MID_VLAN                         active    Fa0/3
30   RIGHT_VLAN                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
```

Switch4

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
```

TRUNK ports between Switches:

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Copy Paste

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up


Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
```

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#cong t
          ^
% Invalid input detected at '^' marker.

Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
```

Switch4

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/4
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch(config-if)#exit
```

DHCP SERVER:

Server0

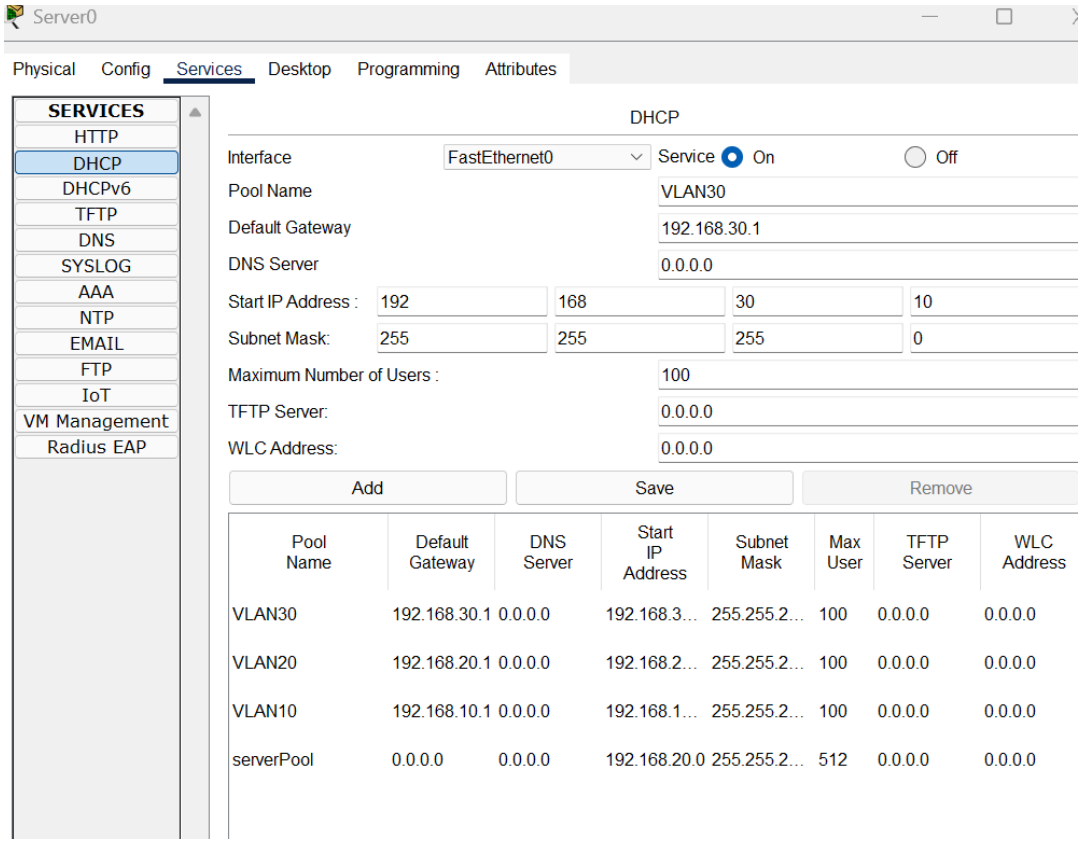
Physical Config Services Desktop Programming Attributes

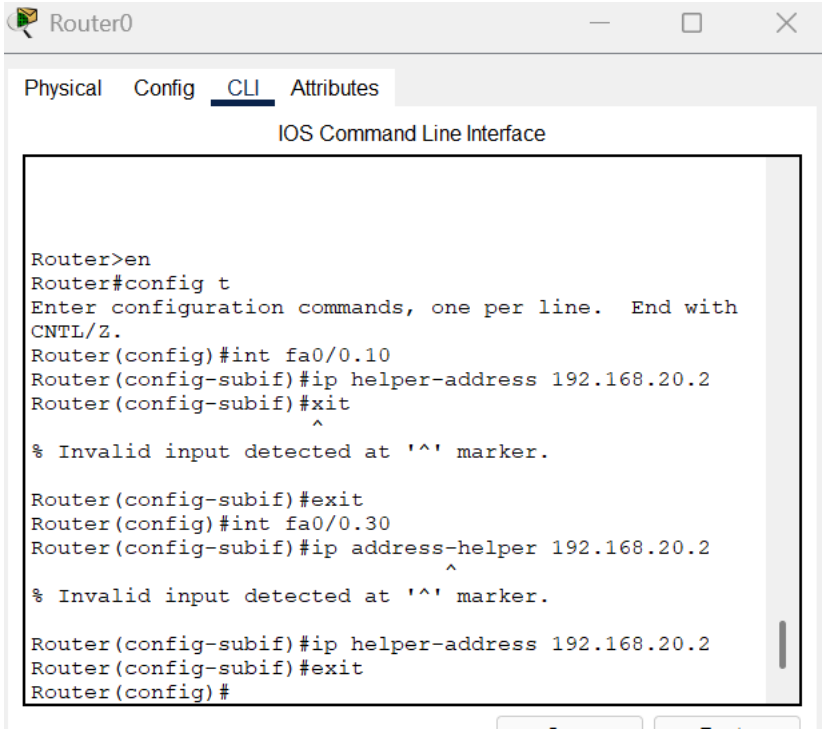
IP Configuration X

IP Configuration

DHCP	Static
IPv4 Address	192.168.20.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0

Creating 3 separate DHCP pools on the server, one for each VLAN, so that each VLAN receives IPs from its correct subnet: VLAN 10 pool, VLAN 20 pool, and VLAN 30 pool.





As we have only one DHCP server connected to VLAN 20, PCs in VLAN 20 can get IP addresses directly from it but for the other VLANs (VLAN 10 and VLAN 30), the DHCP requests need to be forwarded to the server using the IP helper-address command on the router interfaces.

DYNAMIC IP CONFIGURATIONS:

VLAN 10:

PC3

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static DHCP request successful.

IPv4 Address	192.168.10.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

PC2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static DHCP request successful.

IPv4 Address	192.168.10.12
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

IPv6 Configuration

PC0

Physical Config Desktop Programming Attributes

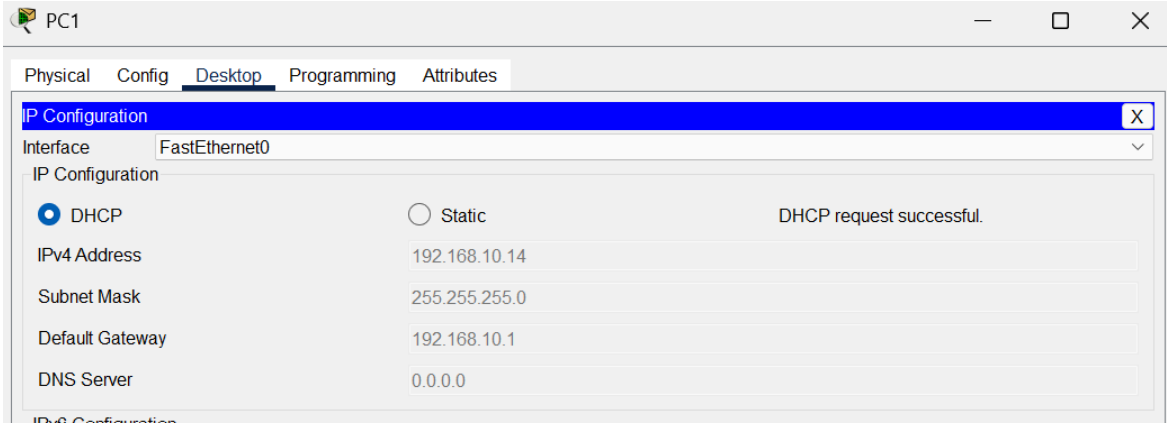
IP Configuration

Interface FastEthernet0

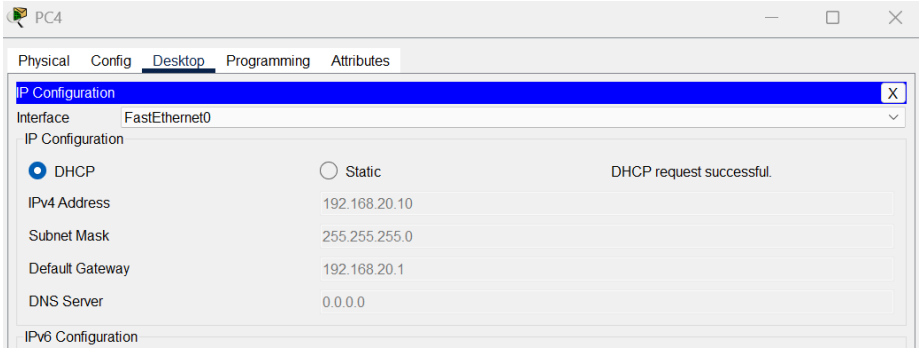
IP Configuration

DHCP Static DHCP request successful.

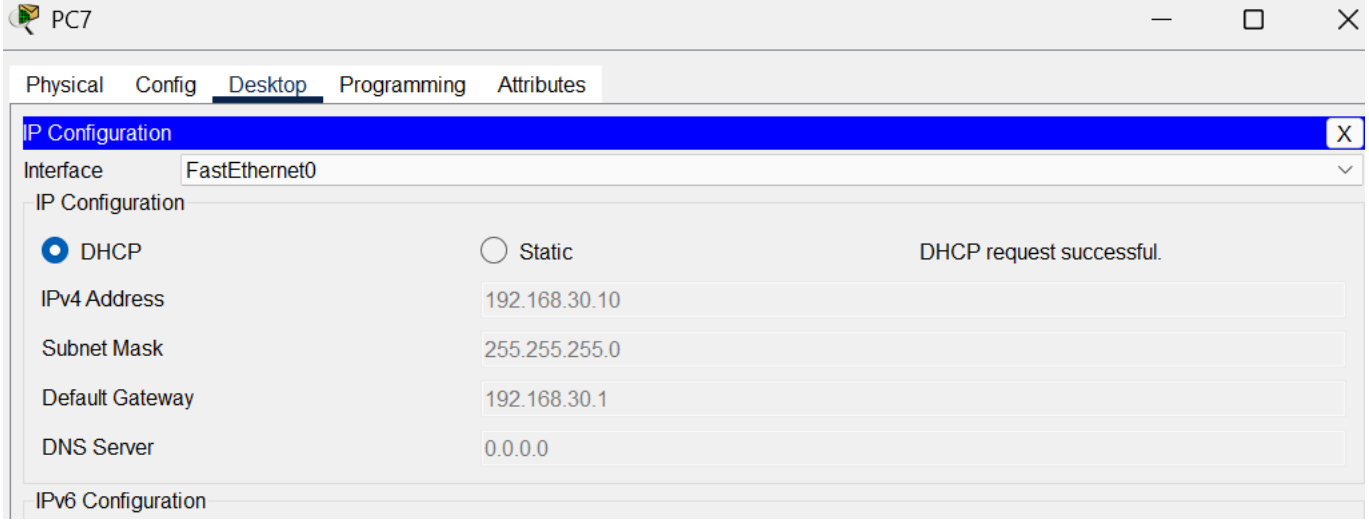
IPv4 Address	192.168.10.13
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

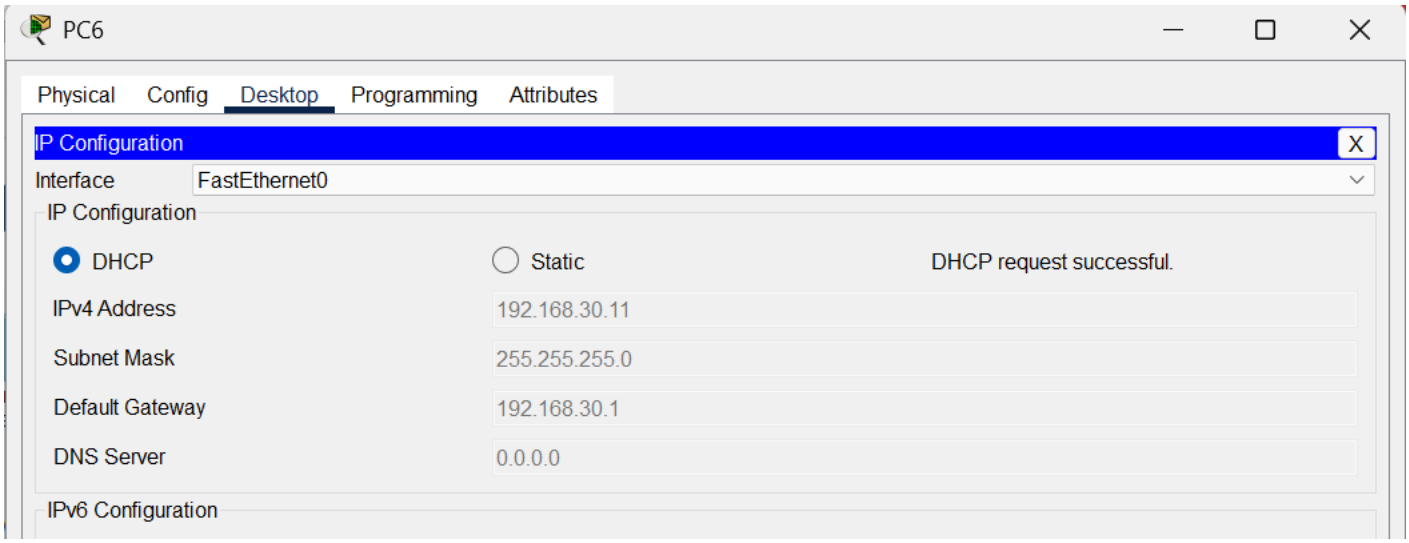


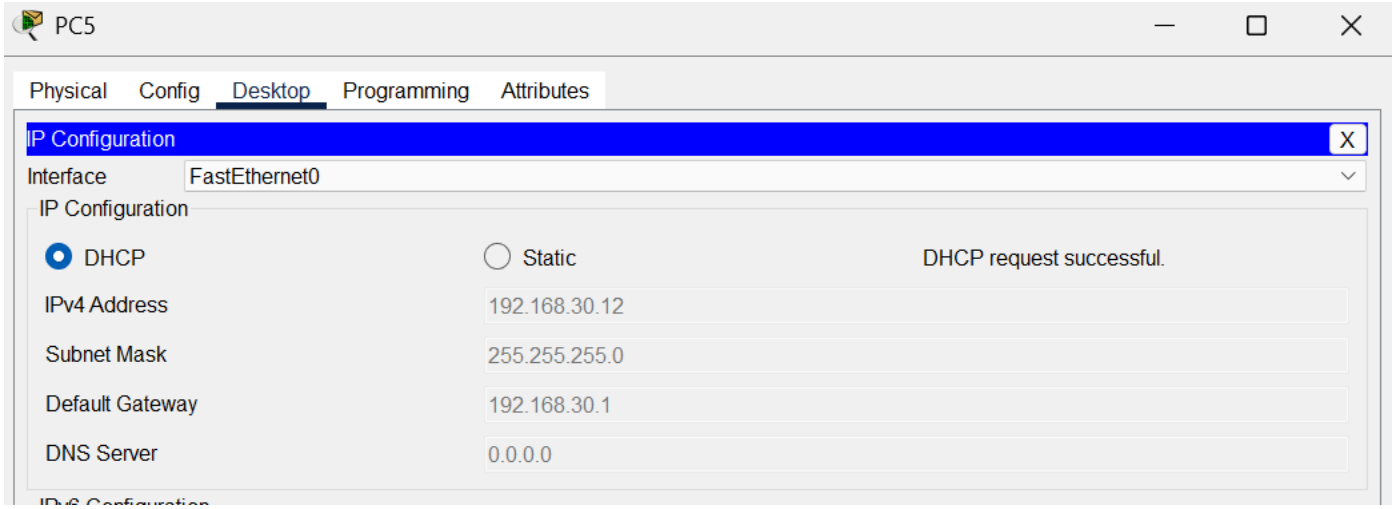
VLAN 20:



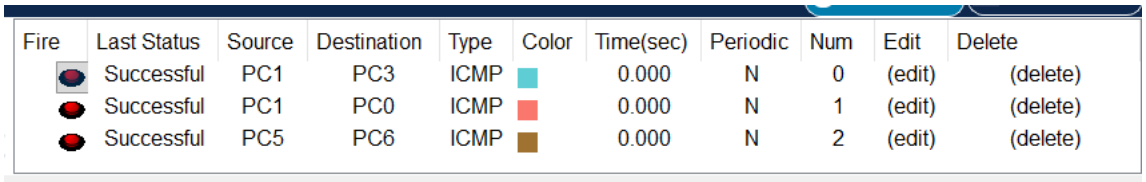
VLAN 30:





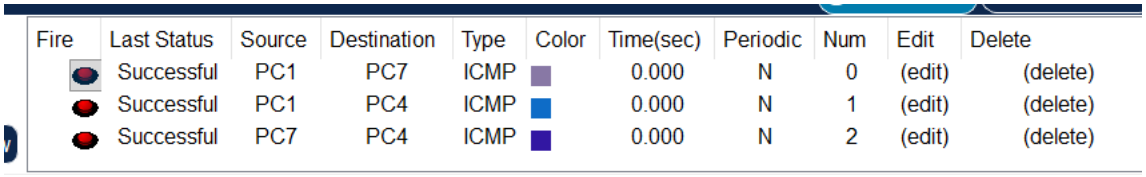


SAME VLAN COMMUNICATION:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC0	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC5	PC6	ICMP		0.000	N	2	(edit)	(delete)

INTER-VLAN COMMUNICATION:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC7	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC7	PC4	ICMP		0.000	N	2	(edit)	(delete)